Mock Paper A(2)

Started on	Thursday, 21 June 2018, 8:30 AM
State	Finished
Completed on	Thursday, 21 June 2018, 8:31 AM
Time taken	10 secs
Grade	**0.00** out of 100.00
Feedback	You are on the right track. Could you spend some more time on your revision?

Question 1
Not answered
Marked out of 1.00
Flag question

MoK Applied Neuroscience 031
A man is unable to mimic the action of hammering a nail as instructed by the examiner. He is exhibiting:

Select one:
- Constructional apraxia
- Finger agnosia
- Ideomotor apraxia
- Dressing apraxia
- Visual object agnosia

Your answer is incorrect.

Ideomotor apraxia is a disorder of goal-directed movement. The patient knows what to do but not how to do it. Tool use is more affected than spontaneous actions/gestures. There is an impairment of pantomiming ability.

The correct answer is: Ideomotor apraxia

Question 2
Not answered
Marked out of 1.00
Flag question

MoK Applied Neuroscience 032
Which of the following tests of WAIS is resistant to brain damage?

Select one:
- Digit span
- Similarities
- Digit symbol
- Block design
- Information

Your answer is incorrect.

The WAIS is composed of 11 subtests made up of six verbal subtests and five performance subtests, which yield a verbal IQ, a performance IQ, and a combined or full-scale IQ. In WAIS, hold tests are vocabulary, information, object assembly and picture completion. Non hold tests are block design, digit span, similarities and digit symbol. A deterioration quotient is derived from the difference between 'don't hold' and 'hold' test scores. Hold tests are supposed to be resistant to age-related decline; so they may be sensitive to organic brain damage such as dementia. But this is disputed.

The correct answer is: Information

Question 3
Not answered
Marked out of 1.00
Flag question

MoK Applied Neuroscience 033
Which of the following statements is true about neural correlates of sign language in deaf patients?

Select one:
- Occipital lobe takes over language processing functions in deaf patients
- Comprehension of sign language does not require involvement of language areas in brain
- Right sided dominance for language is seen in most sign language users
- Left temporal lesions can produce comprehension deficits in sign language users
- Aphasias are not seen in sign language users

Your answer is incorrect.

Deaf patients show a neural organisation of language cortex that is similiar to the non-deaf population. Left temporal cortex is important for comprehending sign language while Broca's area serves in the production of sign language.

The correct answer is: Left temporal lesions can produce comprehension deficits in sign language users

Question 4
Not answered
Marked out of 1.00
Flag question

MoK Applied Neuroscience 034
Damage to which of the following regions unilaterally will produce an upper quadrantanopia?

Select one:
- Superior frontal gyrus
- Medial temporal lobe
- Inferior Parietal Lobule
- Lower bank of calcarine fissure

- Parietal operculum

Your answer is incorrect.

The primary visual cortex lies on the banks of the calcarine fissure. Neurons located on the lower bank are connected to the lateral geniculate nucleus carrying visual information from the nasal or temporal upper visual fields. A lesion of this region can induce an upper quadrantanopia (i.e., loss of one-quarter of the visual field).

The correct answer is: Lower bank of calcarine fissure

Question 5

Not answered

Marked out of 1.00

Flag question

MoK Applied Neuroscience 035

A patient has spent some time in hospital, and the team asked you to review him as his IQ seems to have deteriorated. The best test to measure the patient's premorbid IQ is

Select one:

- Beck's inventory
- Paired associate learning test
- Rey Osterrieth test
- Luria motor test
- National Adult Reading Test

Your answer is incorrect.

The National Adult Reading Test (NART) is a commonly used clinical method to estimate premorbid intelligence levels in English-speaking patients with dementia and other organic brain lesions. It was developed by Hazel Nelson in 1982.

The correct answer is: National Adult Reading Test

Question 6

Not answered

Marked out of 1.00

Flag question

MoK Applied Neuroscience 036

Broca's aphasia is also known as

Select one:

- Fluent aphasia
- Non-fluent aphasia
- Conduction aphasia
- Nominal aphasia
- Jargon aphasia

Your answer is incorrect.

Fluency and repetition are affected while comprehension is preserved in Broca's aphasia.

The correct answer is: Non-fluent aphasia

Question 7

Not answered

Marked out of 1.00

Flag question

MoK Applied Neuroscience 037

Which of the following cognitive tests is useful to check semantic memory?

Select one:

- Copying pentagons and colouring them
- Category fluency and digit-symbol test
- Serial subtraction and proverb interpretation
- Object naming and defining the meaning of words
- Letter fluency

Your answer is incorrect.

Semantic memory involves memory for word meaning and general knowledge, the key neural substrate being the anterior temporal lobe. The bedside tests used include object naming and defining the meaning of words. Patients with semantic breakdown typically complain of loss of words. Vocabulary diminishes, and patients substitute object names with stock-words e.g. "thing". There is a parallel impairment in appreciating the meaning of individual words, which first involves infrequent or unusual words.

The correct answer is: Object naming and defining the meaning of words

Question 8

Not answered

Marked out of 1.00

Flag question

MoK Applied Neuroscience 038

Which one among the following is a neuropsychological theory of ADHD?

Select one:

- Low central coherence
- Reversal learning
- Orbitofrontal striatal executive dysfunction
- Selective attention bias
- Poor external locus of control

Your answer is incorrect.

Attention Deficit Hyperactivity Disorder has been implicated in dysfunction of neural reward circuitry controlling motivation, reward, and impulsivity, including orbitofrontal cortex systems. This condition has increasingly been defined as a disorder resulting from impaired behavioral inhibition leading to executive function deficits.

The correct answer is: Orbitofrontal striatal executive dysfunction

Question 9
Not answered
Marked out of 1.00
Flag question

MoK Applied Neuroscience 039
Lesion of which lobe causes nominal aphasia?

Select one:
- Frontal
- Cerebellum
- Parietal
- Occipital
- Temporal

Your answer is incorrect.

Although many experts have believed that damage to either Broca's or Wernicke's area is the main cause of anomia, current studies have shown that damage in the left parietal lobe is the epicenter of anomic aphasia. (Ref: Fridriksson J, Kjartansson O, Morgan PS, et al. (2010). Impaired speech repetition and left parietal lobe damage. J. Neurosci. 30 (33): 11057-61.)

The correct answer is: Parietal

Question 10
Not answered
Marked out of 1.00
Flag question

MoK Applied Neuroscience 040
Jane experiences sensory aura with tingling, numbness with vague pain sensations, body image distortions, vertigo and visual illusions. The most likely seat of epileptic focus is

Select one:
- Brain stem
- Frontal lobe
- Temporal lobe
- Parietal lobe
- Occipital lobe

Your answer is incorrect.

Pain is rare as an epileptic phenomenon, but one-quarter of parietal lobe seizures have pain as prominent symptom. Somatosensory seizures, somatic illusions, visual distortion, vertigo and body-image distortions are other symptoms of parietal seizures. Ref: http://www.med.nyu.edu/cec/epilepsy/types/parietal_lobe.html

The correct answer is: Parietal lobe

Question 11
Not answered
Marked out of 1.00
Flag question

MoK Applied Neuroscience 044
A lesion in which of the following sites is known to result in bitemporal hemianopia?

Select one:
- Frontal lobe tumour
- Pituitary tumour
- Temporal lobe tumour
- Parietal lobe tumour
- Occipital lobe tumour

Your answer is incorrect.

While most children with pituitary tumours present with the clinical picture of raised intracranial pressure,the majority o f the adults present with visual failure and optic atrophy. Bitemporal hemianopia is fairly frequently found (one-third of patients at the time of diagnosis) but is usually asymmetrical and unpredictable in its evolution. Br J Ophthalmol. 1975 Dec;59(12):689-95.

The correct answer is: Pituitary tumour

Question 12
Not answered
Marked out of 1.00
Flag question

MoK Applied Neuroscience 045
Alexia without agraphia (pure alexia) is seen in lesions of

Select one:
- Middle meningeal artery
- Posterior cerebral territory
- Anterior cerebral territory
- Middles cerebral territory
- Posterior inferior cerebellar artery

Your answer is incorrect.

Alexia without agraphia almost always involves an infarct to the left posterior cerebral artery affecting splenium of the corpus callosum and left visual cortex. So the affected person, who is still able to see with right visual cortex, cannot undertake lexical word processing making him unable to read. But he can still write normally.

The correct answer is: Posterior cerebral territory

Question 13
Not answered
Marked out of 1.00
Flag question

MoK Applied Neuroscience 046
A major use of Glasgow Coma Scale is to

Select one:
- Predict mortality
- Predict development of late complications

- Predict time course for recovery
- Predict development of co-morbidities
- Predict development of early complications

Your answer is incorrect.

The Glasgow Coma Scale (GCS) is routinely used in the acute care setting after traumatic brain injury (TBI) to guide decisions in triage, based on its ability to predict mortality. Although the GCS has been previously demonstrated to predict mortality, efficacy in the prediction of functional outcome is not well established. GCS less than 8 can mean a mortality rate as high as 70%. Balestreri, M., Czosnyka, M., Chatfield, D., Steiner, L., Schmidt, E., Smielewski, P., et al. (2004). Predictive value of Glasgow Coma Scale after brain trauma: Change in trend over the past ten years. Journal of Neurology, Neurosurgery, and Psychiatry, 75(1), 161?162.

The correct answer is: Predict mortality

Question 14

Not answered

Marked out of 1.00

Flag question

MoK Applied Neuroscience 047
A 43-year-old man can speak normally and comprehend what is spoken. He can also write spontaneously and to dictation, but reading comprehension is impaired.

Select one:

- Pure word deafness
- Jargon aphasia
- Broca's aphasia
- Pure word blindness
- Wernicke's aphasia

Your answer is incorrect.

Pure word blindness (alexia but no agraphia or agnosic alexia) is one form of alexia where patients suffer from severe reading problems while other language-related skills such as naming, oral repetition, auditory comprehension or writing are typically intact.

The correct answer is: Pure word blindness

Question 15

Not answered

Marked out of 1.00

Flag question

MoK Applied Neuroscience 049
Immediate repetition of 3 items in MMSE is used to test

Select one:

- Recognition
- Recall
- Digit span
- Fluency
- Registration

Your answer is incorrect.

Delayed recall can be tested by asking to repeat the items after a distraction test

The correct answer is: Registration

Question 16

Not answered

Marked out of 1.00

Flag question

MoK Molecular Genetics 016
Alzheimer's disease is strongly associated with:

Select one:

- Presenile-1 gene on chromosome 1
- Presenilin-2 gene on chromosome 2
- COMT gene on chromosome 21
- CAG repeats on chromosome 4
- APO E4 on chromosome 19

Your answer is incorrect.

ApoE4 on Chr 19 is associated with amyloid precursor protein

The correct answer is: APO E4 on chromosome 19

Question 17

Not answered

Marked out of 1.00

Flag question

MoK Molecular Genetics 035
Which of the following is caused by trinucleotide repeat expansion?

Select one:

- Down's syndrome
- Huntington's disease
- Parkinson's disease
- Schizophrenia
- Lewy body dementia

Your answer is incorrect.

Trinucleotide repeat expansion is the underlying cause for fragile X syndrome and Huntington's disease.

The correct answer is: Huntington's disease

Question 18
Not answered
Marked out of 1.00
Flag question

MoK Molecular Genetics 039
Which of the following components of transcribed RNA is removed before translation occurs?

Select one:
- Introns
- Ribosomal attachment sites
- Cap of RNA
- Aminoacid coding portions
- Tail of RNA

Your answer is incorrect.

An intron is any nucleotide sequence within a gene that is removed before generating the final mature RNA product of a gene. Introns are removed by a process called splicing before protein translation begins. The ribonucleoprotein assembly that removes the intervening sequences from pre-mRNAs through splicing is called the spliceosome.

The correct answer is: Introns

Question 19
Not answered
Marked out of 1.00
Flag question

MoK Molecular Genetics 040
Which of the following genetic element is implicated in Huntington's disease?

Select one:
- Presenilin-1 gene on chromosome 14
- IT-15 chromosome 4
- IT-4 on chromosome 15
- APO E4 on chromosome 19
- IT-1 gene on chromosome 1

Your answer is incorrect.

The mutation in the IT-15 gene ("interesting transcript 15") on chromosome 4 with an unstable region of the harbouring an expanded trinucleotide repeat motif, is associated with Huntington's disease.

The correct answer is: IT-15 chromosome 4

Question 20
Not answered
Marked out of 1.00
Flag question

MoK Molecular Genetics 041
The term used for purpose to identify locus of a gene on the chromosome by studying the extent to which it, co-segregates with a marker gene is called as;

Select one:
- Mutagenesis
- Physical mapping
- Association studies
- Linkage analysis
- Polymerase chain reaction

Your answer is incorrect.

Linkage analysis is used to obtain a coarse chromosomal location of the genes associated with a phenotype under investigation, e.g. a genetic disease or an important quantitative trait

The correct answer is: Linkage analysis

Question 21
Not answered
Marked out of 1.00
Flag question

MoK Molecular Genetics 042
Two parents are both affected with the same autosomal recessive disorder give birth to a clinically normal child. Which of the following can explain the above?

Select one:
- Incomplete penetrance
- Variable expressivity
- Pleiotropy
- Locus heterogeneity
- Allelic heterogeneity

Your answer is incorrect.

Locus heterogeneity refers to mutations at several different loci producing identical phenotypes. A good example is presenilin1, 2 and Beta APP mutations all causing phenotypically similar presenile Alzheimer's dementia. In this example, each parent may have had a different mutation, so the child is a heterozygote carrier at both loci.

The correct answer is: Locus heterogeneity

Question 22
Not answered
Marked out of 1.00
Flag question

MoK Molecular Genetics 043
The term used to describe a point mutation that leads to failure of expression of alleles is

Select one:
- Dominant negative mutation

- Lethal mutation
- Apoptotic function
- Loss of function
- Lesion mutation

Your answer is incorrect.

Loss-of-function mutations are the result of gene product having less or no function. When the allele has a complete loss of function, it is often called an amorphic mutation. Phenotypes associated with such mutations are most often recessive (Mutation - DNA Wiki, http://dnawiki.wikispot.org/Mutation).

The correct answer is: Loss of function

Question 23

Not answered

Marked out of 1.00

Flag question

MoK Molecular Genetics 044
Following a Robertsonian translocation in phenotypically normal individual which of the following types of chromosomes is produced?

Select one:

- Holocentric
- acrocentric
- Metacentric
- telocentric
- submetacentric

Your answer is incorrect.

Following a Robertsonian translocation, the small 'p' arms are discarded, and a metacentric fusion chromosome results.

The correct answer is: Metacentric

Question 24

Not answered

Marked out of 1.00

Flag question

MoK Molecular Genetics 045
A family with three affected children has a higher recurrence risk for a disease than another family with only two affected children. This disease is most likely to be inherited via

Select one:

- Mitochondrial
- Autosomal dominant
- Autosomal recessive
- Multifactorial
- X linked dominant

Your answer is incorrect.

An increasing recurrence risk according to the number of relatives affected is seen in multifactorial diseases. Recurrence risks for Mendelian disorders are unaffected by number of offsprings already affected.

The correct answer is: Multifactorial

Question 25

Not answered

Marked out of 1.00

Flag question

MoK Molecular Genetics 046
Which of the following disorders is associated with HLA-DQB1*-0602?

Select one:

- Huntington's disease
- Narcolepsy
- Pick's disease
- Schizophrenia
- Bipolar disorder

Your answer is incorrect.

There is a strong association between HLA-DR2 haplotypes coded on chromosome 6 and narcolepsy. HLA-DQB1*0602 and DQA *0102 are found in 85-95% of individuals, compared to 12-38% in the general population.

The correct answer is: Narcolepsy

Question 26

Not answered

Marked out of 1.00

Flag question

MoK Molecular Genetics 047
A patient with schizophrenia is less likely to get married and produce children due to the nature of the illness. This will serve to reduce the gene pool of schizophrenia causing genes. This is called as

Select one:

- Pleiotropy
- Natural selection
- Assortative mating
- Genetic drift
- Negative eugenics

Your answer is incorrect.

Natural selection serves to maintain a low pool of certain disorders. Due to impaired fertility and affliction at reproductive age, schizophrenia genes may be maintained at a lower level of constancy in the population.

The correct answer is: Natural selection

Question 27
Not answered
Marked out of 1.00
Flag question

MoK Neuroanatomy 012
A man is brought to the A& E department with akinetic mutism. He has suffered a stroke. Which artery is most likely to be involved?

Select one:
- Basilar artery
- Subclavian artery
- Posterior inferior cerebellar artery
- Middle cerebral artery
- Bilateral anterior cerebral artery

Your answer is incorrect.

The cingulate gyrus is supplied bilaterally by the anterior cerebral artery. An occlusion involving bilateral anterior cerebral arteries could damage the cingulate gyri, leading to an akinetic mute state.

The correct answer is: Bilateral anterior cerebral artery

Question 28
Not answered
Marked out of 1.00
Flag question

MoK Neuroanatomy 016
Communication between the frontal lobes of each side of the brain is mediated through which of the following structures?

Select one:
- Fornix
- Genu of corpus callosum
- Parahippocampal gyrus
- Splenium of corpus callosum
- Arcuate fasciculus

Your answer is incorrect.

The corpus callosum forms the major connection by which the hemispheres communicate with one another. The genu of the corpus callosum contains fibers that pass from the frontal lobe of one side to that of the other.

The correct answer is: Genu of corpus callosum

Question 29
Not answered
Marked out of 1.00
Flag question

MoK Neuroanatomy 018
Which of the following structures appears larger on the left side than the right in healthy right-handed man?

Select one:
- Frontal lobe
- Hippocampus
- Inferior temporal gyrus
- Caudate nucleus
- Heschl's gyrus

Your answer is incorrect.

In normal healthy right-handed subjects, Left larger than right symmetry is seen with structures related to language processing such as planum temporale and Heschl's gyrus (primary auditory cortex)

The correct answer is: Heschl's gyrus

Question 30
Not answered
Marked out of 1.00
Flag question

MoK Neuroanatomy 019
A 40-year-old man develops depressed mood, anhedonia, initial and terminal insomnia, loss of appetite, a 10-lb weight loss, and difficulty with sexual arousal. The clinical features of the patient's psychiatric illness suggest dysfunction of the

Select one:
- Hippocampus
- Corpus callosum
- Frontal lobes
- Pituitary
- Hypothalamus

Your answer is incorrect.

Clinical studies of patients with major depressive disorders indicate that an intrinsic regulatory defect involving the hypothalamus underlies the disorder. It also involves the monoamine pathways. The hypothalamic modulation of neuroendocrine activity has been implicated, as have been the neurotransmitter systems of serotonin and norepinephrine.

The correct answer is: Hypothalamus

Question 31
Not answered

MoK Neuroanatomy 020
Which of the following statements about microglia is correct?

Marked out of 1.00

Flag question

Select one:

- It is mainly responsible for myelination
- It constitutes 70% of glial population in brain
- It is sensitive to sodium channel stimulation
- It is mesodermal in origin
- It is present outside brain and spinal cord

Your answer is incorrect.

Some authors claim that microglia originate from the neuroepithelium, although most now believe that microglial cells are of mesodermal origin, and probably belong to the monocyte/macrophage cell line. These cells must enter the developing central nervous system (CNS) from the blood stream, the ventricular space or the meninges. Afterward, microglial cells are distributed more or less homogeneously through the entire nervous parenchyma. Prog Neurobiol. 1998 Oct;56(2):173-89.

The correct answer is: It is mesodermal in origin

Question 32

Not answered

Marked out of 1.00

Flag question

MoK Neurochemistry 016

Which of the following is the primary mechanism involved in removal of the transmitter at the neuromuscular junction?

Select one:

- Reuptake
- Enzymatic degradation
- Diffusion
- Immune complex formation
- Renal clearance

Your answer is incorrect.

A transmitter can be removed from the synaptic cleft through one of the following three modes: (1) enzymatic degradation, (2) reuptake, and (3) diffusion. In neuromuscular junctions acetylcholine, the primary neurotransmitter, undergoes enzymatic degradation.

The correct answer is: Enzymatic degradation

Question 33

Not answered

Marked out of 1.00

Flag question

MoK Neurochemistry 017

Removal of noradrenaline from the region of the synaptic cleft may be achieved by which of the following mechanisms?

Select one:

- reuptake only
- diffusion only
- conversion to adrenaline
- enzymatic degradation only
- enzymatic degradation, diffusion, and reuptake

Your answer is incorrect.

The most common of these mechanisms is reuptake, in which mediated by high-affinity transmitter-specific transporter molecules.

The correct answer is: enzymatic degradation, diffusion, and reuptake

Question 34

Not answered

Marked out of 1.00

Flag question

MoK Neurochemistry 018

The metabotropic receptors are coupled to

Select one:

- Ligand non-dependent regulators of nuclear transcription
- G proteins
- Ligand dependent regulators of nuclear transcription
- Proteins that have intrinsic enzyme activity
- Ligand gated channels

Your answer is incorrect.

Receptors are categorized into: (1) ionotropic: ligand-gated channels, in which binding of a chemical messenger alters the probability of opening of transmembrane pores or channels; (2) metabotropic: those in which the receptor proteins are coupled to intracellular G proteins as transducing elements (3) intrinsic enzymes: those consisting of single membrane-spanning protein units that have intrinsic enzyme activity (for example, having tyrosine kinase activity; some authors classify them alongside metabotropic receptors); and (4) ligand-dependent regulators of nuclear transcription (including receptors for corticosteroids such as testosterone)

The correct answer is: G proteins

Question 35

Not answered

Marked out of 1.00

Flag question

MoK Neurochemistry 019

The neurotransmitter involved in the long-term potentiation of memory is

Select one:

- Noradrenaline
- Dopamine
- Acetyl choline
- Glutamate

- Serotonin

Your answer is incorrect.

Long-term potentiation is a long-lasting enhancement of signal transmission between 2 neurotransmitters that results from stimulating them synchronously. It is considered as one of the major cellular mechanisms that underlie learning and memory. Glutamate, the major excitatory neurotransmitter has NMDA receptors which mediate long-term potentiation.

The correct answer is: Glutamate

Question 36

Not answered

Marked out of 1.00

Flag question

MoK Neurochemistry 020

The neurotransmitter which is most widely distributed in the brain is

Select one:

- Dopamine
- Glutamate
- Acetylcholine
- Norepinephrine
- Serotonin

Your answer is incorrect.

The most prevalent neurotransmitter in the brain is glutamate. It is the major excitatory neurotransmitter with wide distribution in cerebral cortex, hippocampus and neocortical projections in the brain

The correct answer is: Glutamate

Question 37

Not answered

Marked out of 1.00

Flag question

MoK Neuropathology 010

In a patient being evaluated for neuropsychiatric disturbances, CT scan shows atrophy of head of caudate nucleus. Which of the following is a likely diagnosis?

Select one:

- Alzheimer's disease
- Wilson's disease
- Fahr's disease
- Huntington's disease
- Pick's disease

Your answer is incorrect.

Huntington's is characterised by this finding especially in late stages; in Fahr's disease calcified basal ganglia are noted.

The correct answer is: Huntington's disease

Question 38

Not answered

Marked out of 1.00

Flag question

MoK Neuropathology 011

Which of the following neuroimaging finding is most suggestive of new variant CJD?

Select one:

- Occipital hypoperfusion on HMPAO- SPECT
- Activation of primary auditory cortex on functional MRI
- Impaired dorsolateral cortex function
- Medial temporal atrophy on structural MRI scan
- Increased signal in the pulvinar nucleus of thalamus bilaterally

Your answer is incorrect.

The MRI is the most useful supportive diagnostic test in variant CJD. A characteristic abnormality seen in the posterior thalamic region (pulvinar sign) is highly sensitive and specific for variant CJD.

The correct answer is: Increased signal in the pulvinar nucleus of thalamus bilaterally

Question 39

Not answered

Marked out of 1.00

Flag question

MoK Neuropathology 012

A patient develops viral encephalitis. He exhibits increased oral exploratory behaviour and inappropriate sexuality. This syndrome can be called

Select one:

- Kluver Bucy syndrome
- Korsakoff syndrome
- Lennox Gastaut syndrome
- Gerstmann syndrome
- Klein Levine syndrome

Your answer is incorrect.

Hypermetamorphopsia (compulsion to attend to all visual stimuli), hyperorality, hypersexuality, blunted emotional reactivity is called Kluver-Bucy syndrome; the full syndrome is rarely seen in man, but partial syndrome can occur with viral encephalitis affecting temporal lobe.

The correct answer is: Kluver Bucy syndrome

Question 40
Not answered
Marked out of 1.00
Flag question

MoK Neurophysiology 010
A 44 year old man is treated for long standing asthma with a combination of oral theophylline, prednisolone, inhaled salbutamol and inhaled budesonide. He appears heavy with purplish striae on abdomen and excessive fat on the nape of his neck. His asthma is under control but he exhibits behaviours and mental state consistent with hypomania. Which of the following is a possible cause?

Select one:
- Late onset bipolar disorder
- Theophylline induced epileptiform activity
- Hypothyroidism
- Cushing's syndrome
- Secondary diabetes mellitus

Your answer is incorrect.
The story is consistent with Cushing's syndrome. It is most likely due to oral prednisolone, not inhaled budesonide.
The correct answer is: Cushing's syndrome

Question 41
Not answered
Marked out of 1.00
Flag question

MoK Neurophysiology 011
During EEG recording a resting subject is asked to focus his attention on a cognitive task. Which of the following will be observed?

Select one:
- No change in resting EEG
- Unilateral change in resting EEG at dominant hemisphere
- Appearance of alpha waves on focussing attention
- Disappearance of alpha waves on focussing attention
- Disappearance of theta waves on focussing attention

Your answer is incorrect.
The disappearance of alpha waves on focussing attention is called as an alpha block; this occurs when attention is directed.(Ganong's Review of Physiology 21e, Chapter 11., Lange).
The correct answer is: Disappearance of alpha waves on focussing attention

Question 42
Not answered
Marked out of 1.00
Flag question

MoK Neurophysiology 012
In the dementia associated with Alzheimer's disease, the EEG will usually show which of the following?

Select one:
- Periodic frontal lobe discharges
- Generalized background slowing
- Focal slowing
- An isoelectric record
- Spike-and-wave discharges

Your answer is incorrect.
In Alzheimer's disease, the frequency of background alpha rhythm may slow down with reduced duration. Periodic discharges in the form of sharp waves or spikes may develop during Creutzfeldt-Jakob disease. EEG is otherwise not especially helpful in distinguishing between the common causes of dementia.
The correct answer is: Generalized background slowing

Question 43
Not answered
Marked out of 1.00
Flag question

MoK Adverse Effects 031
Which of the following drugs can cause whitish-brown granular deposits in the cornea and lens ?

Select one:
- Thioridazine
- Amisulpride
- Chlorpromazine
- Quetiapine
- Olanzapine

Your answer is incorrect.
Irreversible pigmentation (similar to retinitis pigmentosa) with thioridazine, whitish-brown granular deposits in the cornea and lens with chlorpromazine, and rarely glaucoma (and other eye changes) with quetiapine are some of the ocular side effects of neuroleptics.
The correct answer is: Chlorpromazine

Question 44
Not answered
Marked out of 1.00
Flag question

MoK Adverse Effects 038
Risk factors for tardive dyskinesia includes all of the following except

Select one:
- Using high potency drugs
- Having affective disorder diagnosis
- Continuous use without drug holidays

- Being Afro Caribbean
- Having neurological damage

Your answer is incorrect.

Drug 'holidays' are associated with more frequent relapses, leading to higher than usual doses of antipsychotics being used cumulatively This is associated with higher risk of Tardive Dyskinesia.

The correct answer is: Continuous use without drug holidays

Question 45
Not answered
Marked out of 1.00
Flag question

MoK Adverse Effects 039
Rabbit syndrome presenting with mouth tremors is a result of

Select one:
- D1 receptor blockade
- 5HT2C receptor blockade
- D2 receptor blockade
- 5HT2A receptor blockade
- D4 receptor blockade

Your answer is incorrect.

The rabbit syndrome is a tremor affecting the lips and perioral muscles and occurs late in the course of antipsychotic treatment. It is considered as a form of extrapyramidal symptom related to D2 receptor blockade.

The correct answer is: D2 receptor blockade

Question 46
Not answered
Marked out of 1.00
Flag question

MoK Adverse Effects 040
A patient with schizophrenia has been functioning considerably better since he started taking clozapine. Currently he is on a dose of 600 mg/day, but he has experienced an episode of new onset seizures. Which of the following is the most appropriate next step?

Select one:
- Add carbamazepine to clozapine
- Add valproate to the same dose of clozapine
- Add amisulpride
- Decrease the dose of clozapine
- Stop clozapine for 6 weeks

Your answer is incorrect.

Seizures related to clozapine are dose-/dose increase-related. In those patients who experience seizures on clozapine, prophylactic valproate can be considered if on a high dose or with high plasma level (more than 500 ng/l). After a seizure, withhold clozapine for one day; restart at reduced dose; give sodium valproate. EEG abnormalities are common in those on clozapine and does not warrant alteration in the treatment automatically. (Maudsley Prescribing Guidelines 9thEd)

The correct answer is: Decrease the dose of clozapine

Question 47
Not answered
Marked out of 1.00
Flag question

MoK Adverse Effects 041
Benzodiazepine withdrawal is associated with:

Select one:
- Increased brain GABA function
- Decreased brain dopamine function
- Decreased brain GABA function
- Increased brain serotonin function
- Increased brain dopamine function

Your answer is incorrect.

People become tolerant to benzodiazepines probably because their nerve cells respond by producing fewer receptors for GABA/benzodiazepines. This phenomenon, known as 'down-regulation', means that the number of 'high affinity' GABA receptors decreases in response to the enhancement of GABA caused by the drug. The body responds to the continued presence of the drug with a series of adjustments that tend to overcome the drug effects. In the case of benzodiazepines, compensatory changes occur in the GABA and benzodiazepine receptors which become less responsive, so that the inhibitory actions of GABA and benzodiazepines are decreased (Ref: CH Ashton. The role of GABA and GABA receptors in benzodiazepine withdrawal. http://www.benzosupport.org/gaba.htm)

The correct answer is: Decreased brain GABA function

Question 48
Not answered
Marked out of 1.00
Flag question

MoK Adverse Effects 042
A 48-year-old gentleman suffers from schizophrenia and is on Olanzapine treatment. He has a BMI of 38 and suffers from drowsiness and erectile dysfunction. His diagnosis would be

Select one:
- Kluver bucy syndrome
- Cushing's syndrome
- Addison's disease
- Klein Levin syndrome
- Diabetes mellitus

Your answer is incorrect.

Erectile dysfunction (ED) is a common problem amongst men who have diabetes affecting 35-75% of male diabetics, which could be induced by Olanzapine. The other side effects of Olanzapine include drowsiness and weight gain. Antipsychotic induced sedation and weight gain may reduce sexual desire.

The correct answer is: Diabetes mellitus

Question 49
Not answered
Marked out of 1.00
Flag question

MoK Adverse Effects 043
Which of the following drugs can markedly increase blood acetaldehyde concentration?

Select one:
- Acamprosatwe
- Morphine
- Disulfiram
- Naltrexone
- Biperiden

Your answer is incorrect.

Aldehyde dehydrogenase (ALDH) is the enzyme that facilitates the catabolism of acetaldehyde to acetic acid. When the action of ALDH is inhibited by disulfiram, acetaldehyde is not converted to acetate but accumulates in the blood. Most of the symptoms of the disulfiram-ethanol are due to the increased circulating acetaldehyde.

The correct answer is: Disulfiram

Question 50
Not answered
Marked out of 1.00
Flag question

MoK Adverse Effects 044
A 33-year-old woman with a history of recurrent depressive disorder and is on fluoxetine 20 mg/day. She wishes to conceive and if she takes fluoxetine during pregnancy, what is the risk of it causing major malformations in the foetuses?

Select one:
- Trebles the risk
- Does not appear to significantly increase the risk
- Doubles the risk
- Quadruples the risk
- More than quadruples the risk

Your answer is incorrect.

Tricyclics and fluoxetine do not have a known teratogenic effect in humans. Miscarriages do not seem to be increased by fluoxetine

The correct answer is: Does not appear to significantly increase the risk

Question 51
Not answered
Marked out of 1.00
Flag question

MoK Adverse Effects 045
The drug of choice for treatment of restless legs syndrome is

Select one:
- Antidepressant
- Dopamine agonist
- Baclofen
- Clonazepam
- Antipsychotic

Your answer is incorrect.

Currently, dopamine agonists such as ropinirole, pramipexole, carbidopa/levodopa or pergolide are commonly used for the treatment of restless legs syndrome. Clonazepam may also be beneficial.

The correct answer is: Dopamine agonist

Question 52
Not answered
Marked out of 1.00
Flag question

MoK Adverse Effects 046
In patients receiving lithium therapy, the following can be observed even in normal therapeutic doses:

Select one:
- ECG changes
- Coarse tremors
- Bradycardia
- Pulmonary oedema
- Ataxia

Your answer is incorrect.

Reversible flattening and inversion of T-waves are seen even at therapeutic doses, but usually produce no clinical consequences. Br. J. Clin. Pharmac. (1980), 9, 599-604

The correct answer is: ECG changes

Question 53
Not answered

MoK Adverse Effects 047
One among the following is not a feature of serotonin syndrome

Marked out of 1.00

Flag question

Select one:

- Hyperreflexia
- Confusion
- Myoclonic Jerks
- Electric shock like sensations
- Autonomic instability

Your answer is incorrect.

Serotonin syndrome: Confusion, pyrexia, sweating, Myoclonic Jerks, Hyperreflexia, Autonomic instability, gastrointestinal symptoms, mood changes and mania. Note that shock-like sensations are seen in SSRI discontinuation syndrome.

The correct answer is: Electric shock like sensations

Question 54

Not answered

Marked out of 1.00

Flag question

MoK Adverse Effects 048
Which of the following is a potentially serious adverse effect associated with tacrine?

Select one:

- Elevated liver enzymes
- Blood dyscrasias
- Respiratory arrest
- Retinal damage
- Renal failure

Your answer is incorrect.

Elevated liver enzymes (potentially serious), anorexia, nausea, rash, myalgia, loss of weight, and increased sweating are some of the side effects of tacrine, the first introduced anti-dementia drug. Donepezil is specific for cholinesterase inhibition in the CNS while tacrine is not. Donepezil, therefore, has less peripheral adverse effects.

The correct answer is: Elevated liver enzymes

Question 55

Not answered

Marked out of 1.00

Flag question

MoK Adverse Effects 050
Which of the following is a predisposing factor for QT prolongation?

Select one:

- Hyperkalemia
- Hypermagnesemia
- Male gender
- Hypercalcemia
- Female gender

Your answer is incorrect.

New cases of long QT syndrome are diagnosed more in female patients (60-70% of cases) than male patients. The female predominance may be related to the relatively prolonged QTc (as determined by using the Bazett formula) in women compared with men and to a relatively higher mortality rate in young men (Long QT Syndrome - Medscape Reference, http://emedicine.medscape.com/article/157826-overview (accessed April 13, 2015).). Electrolyte abnormalities (especially hypokalaemia or hypomagnesaemia) are also associated.

The correct answer is: Female gender

Question 56

Not answered

Marked out of 1.00

Flag question

MoK Pharmacodynamics 021
Which of the following antipsychotics is the fastest to dissociate from the Dopamine D2 receptor?

Select one:

- Aripiprazole
- Xanomeline
- Clozapine
- Paliperidone
- Risperidone

Your answer is incorrect.

Animal data show that a rapid dissociation from the D(2) receptor at a molecular level produces the atypical antipsychotic effect. In vitro data show that the single most powerful predictor of atypicality for the current generation of atypical antipsychotics is fast dissociation from the D(2) receptor, not its high affinity at 5-HT(2), D(4), or another receptor. It has also been postulated that clozapine, as well as quetiapine, bind more loosely to the Dopamine receptor, and show a fast dissociation (Kapur & Seeman, 2001. Retrieved from http://ajp.psychiatryonline.org/doi/abs/10.1176/appi.ajp.158.3.360).

The correct answer is: Clozapine

Question 57

Not answered

Marked out of 1.00

Flag question

MoK Pharmacodynamics 022
Which of the following street drugs has local anaesthetic properties?

Select one:

- Ketamine
- Cannabis
- LSD

- Cocaine
- Heroin

Your answer is incorrect.

Cocaine has membrane stabilising, and thus local anaesthetic effects.

The correct answer is: Cocaine

Question 58
Not answered
Marked out of 1.00
Flag question

MoK Pharmacodynamics 023
A 41-year-old woman is taking disulfiram to aid abstinence from alcohol. She present to A and E with sickness and feeling unwell. On examination she appears to have facial flushing with low BP but high pulse rate. She denies having drunk alcohol. Which of the following might explain her symptoms?

Select one:
- Mature cheese
- 2 mugs of black coffee
- Cough syrup
- Banana yoghurt
- Tramadol tablets

Your answer is incorrect.

Many cough and common cold ailments contain a significant amount of alcohol sufficient to precipitate an alcohol-disulfiram reaction.

The correct answer is: Cough syrup

Question 59
Not answered
Marked out of 1.00
Flag question

MoK Pharmacodynamics 025
The antiemetic property of chlorpromazine is due to

Select one:
- D2 blockade
- 5HT2A blockade
- Antiadrenergic effect
- Anticholinergic effect
- 5HT3 blockade

Your answer is incorrect.

D2 blockade due to chlorpromazine in chemoreceptor trigger zone explains its antiemetic property.

The correct answer is: D2 blockade

Question 60
Not answered
Marked out of 1.00
Flag question

MoK Pharmacodynamics 026
Sally is taking Amisulpride and is complaining of amenorrhea, galactorrhoea and loss of sexual desire. This effect is caused by

Select one:
- 5HT1A antagonism
- D1 blockade
- D2 blockade
- D3 blockade
- 5HT2A antagonism

Your answer is incorrect.

This patient is suffering from clinical features of hyperprolactinemia (amenorrhea, galactorrhoea and loss of sexual desire) caused by antipsychotics. Hyperprolactinemia caused by antipsychotics is due to D2 blockade and/or 5HT1A stimulation. Dopamine inhibits prolactin release and so dopamine antagonists can be expected to increase prolactin levels.

The correct answer is: D2 blockade

Question 61
Not answered
Marked out of 1.00
Flag question

MoK Pharmacodynamics 027
Which of the following is a known psychiatric effect of the drug clonidine?

Select one:
- Psychosis
- Mania
- Anorexia
- Depression
- Dementia

Your answer is incorrect.

Other drugs that can cause depression include propranolol, digoxin, frusemide, reserpine, cimetidine, methyldopa, bromocriptine, and phenothiazines.

The correct answer is: Depression

Question 62
Not answered
Marked out of 1.00
Flag question

MoK Pharmacodynamics 028
The term "intrinsic activity" of a drug-receptor complex refers to

Select one:
- The efficiency of a drug in reaching the appropriate receptors
- Potency of the drug to cause a functional response
- Affinity of the drug to the receptor
- Efficacy of the resulting functional response
- The receptor action in the absence of the drug molecule

Your answer is incorrect.

Intrinsic activity refers to the capability of a drug-receptor complex to produce the highest functional response (i.e. efficacy).

The correct answer is: Efficacy of the resulting functional response

Question 63
Not answered
Marked out of 1.00
Flag question

MoK Pharmacodynamics 029
Gamma hydroxybutyrate (GHB) acts on which of the following receptors?

Select one:
- Dopamine
- Opiod
- GABA-B
- Serotonin
- Noradrenaline

Your answer is incorrect.

GHB is primarily a GABAB agonist, naturally occurring in the brain, as well as in wine, beef, small citrus fruits, and almost all animals in small amounts. It was synthesized as an anesthetic initially; later banned due to abuse by body builders (as it was thought to boost growth hormone release) and use as a date rape drug (colourless liquid, easily concealed). It was also abused for euphoric effects that are short lived. It is used as substitute for ecstasy in dance parties (raves)- called liquid ecstasy

The correct answer is: GABA-B

Question 64
Not answered
Marked out of 1.00
Flag question

MoK Pharmacokinetics 029
A patient who has had a renal transplant is currently on cyclosporine for immunosuppression. He appears to be clinically depressed. The renal physician says it has been difficult titrating the cyclosporine levels. Which of the following drugs must be avoided as the first line intervention?

Select one:
- amitriptyline
- fluoxetine
- venlafaxine
- citalopram
- phenelzine

Your answer is incorrect.

Fluoxetine, fluvoxamine and nefazodone are potent CYP3A4 inhibitors and can significantly affect cyclosporine levels in patients.

The correct answer is: fluoxetine

Question 65
Not answered
Marked out of 1.00
Flag question

MoK Pharmacokinetics 030
A 60-year-old gentleman has a diagnosis of bipolar affective disorder and is on Lithium carbonate. He has recently developed systemic hypertension. Which one among the following is the most appropriate drug to treat his hypertension?

Select one:
- Bendroflumethiazide
- Amiloride
- Captopril
- Furosemide
- Chlorthalidone

Your answer is incorrect.

Diuretics can increase serum lithium levels markedly by decreasing its clearance. Thiazides are the worst culprits while loop diuretics are somewhat safer. ACE Inhibitors decrease the excretion of Lithium and can also precipitate renal failure.

The correct answer is: Furosemide

Question 66
Not answered
Marked out of 1.00
Flag question

MoK Pharmacokinetics 031
The safest diuretic to be used with lithium is

Select one:
- Bendrofluomethiazide
- Furosemide
- Xipamide

- Chlorthalidone
- Indapamide

Your answer is incorrect.

The following are loop diuretics- furosemide, bumetanide and torasemide. Although there are case reports of lithium toxicity induced by loop diuretics, many patients receive this combination of drugs without apparent problems. Patients taking blood diuretics may also have been advised to restrict their salt intake, which may contribute to the risk of lithium toxicity in these individuals. Thiazide diuretics (bendroflumethiazide, chlorthalidone, indapamide, metolazone, xipamide), care is required, as these classes of drugs can reduce the renal clearance of lithium and precipitate toxicity. (Ref: The Maudsley prescribing guidelines)

The correct answer is: Furosemide

Question 67

Not answered

Marked out of 1.00

Flag question

MoK Pharmacokinetics 032
A patient on MAOI can consume

Select one:

- Grape fruit juice
- Broad beans
- Red wine
- Pickled herring
- Cheese

Your answer is incorrect.

Unlike the others listed, grapefruit juice is not rich in tyramine

The correct answer is: Grape fruit juice

Question 68

Not answered

Marked out of 1.00

Flag question

MoK Pharmacokinetics 034
Which of the following is true regarding diazepam?

Select one:

- Has 95% protein binding
- Has an inactive metabolite called desmethyldiazepam
- Does not cause blurred vision
- Half-life is 12 hours
- Presence of food increases the extent of absorption

Your answer is incorrect.

Diazepam is a 1,4-benzodiazepine. Diazepam has an active metabolite desmethyldiazepam. It is rapidly absorbed with nearly 100% oral and close to 95% is bound in plasma. Food may inhibit absorption. The half-life of diazepam is 43±13 hours but ranges from 40-100 hours if the contribution from active metabolites is included. Side effects may include dry mouth, blurred or double vision, headache, vertigo, urinary retention, excessive perspiration, nausea and vomiting, ataxia, tremor, depression, hypotension and diminished reflexes (Drugs and Human Performance FACT SHEETS - Diazepam. Retrieved from http://www.nhtsa.gov/People/injury/research/job185drugs/diazepam.htm).

The correct answer is: Has 95% protein binding

Question 69

Not answered

Marked out of 1.00

Flag question

MoK Pharmacokinetics 035
Which one among the following statements about mianserin is true?

Select one:

- Inhibits uptake of both Noradrenaline and 5-HT
- Has considerable alpha 1 adrenoreceptor blocking properties
- Has a plasma half life that allows once daily dosage
- Has strong antimuscarinic effects
- Has significant sedative effects

Your answer is incorrect.

Mianserin is a potent inhibitor of Noradrenaline uptake with little effect on 5-HT. It is an alpha 2 antagonist and has weak alpha 1 antagonism. It is mildly sedative and has weak antimuscarinic effects. It is rapidly absorbed from the gut with the peak plasma concentration appearing in less than 3 hours after a single oral dose. The plasma half-life is 10-20 hours.

The correct answer is: Has a plasma half life that allows once daily dosage

Question 70

Not answered

Marked out of 1.00

Flag question

MoK Pharmacokinetics 036
Which of the following is NOT a phase 2 metabolic reaction in drug elimination?

Select one:

- N acetylation
- Conjugation
- Methylation
- Hydrolysis
- Glucuronidation

Your answer is incorrect.

Conjugation, N-acetylation, methylation and glucuronidation are phase 2 reactions. Oxidation and hydrolysis are phase 1 reactions.

The correct answer is: Hydrolysis

Question 71

Not answered

Marked out of 1.00

Flag question

MoK Adverse Effects 051

Thromboembolism as a possible side effect of clozapine treatment seems most likely to occur in the

Select one:

- Within 6-8 weeks of starting treatment
- First 3 months of treatment
- First 4 days of treatment
- First 4 months of treatment
- First 4 weeks of treatment

Your answer is incorrect.

Clozapine's adverse effect of Thromboembolism may be related to its observed effect on Antiphospholipid antibodies; though rare, if it occurs, this is usually in the first three months of the treatment.

The correct answer is: First 3 months of treatment

Question 72

Not answered

Marked out of 1.00

Flag question

MoK Pharmacokinetics 037

Which of the following antidepressants' plasma levels correlates with therapeutic response?

Select one:

- Imipramine
- Mirtazapine
- Amoxapine
- Sertraline
- Duloxetine

Your answer is incorrect.

The clinical outcome when using imipramine is unequivocally related to the plasma level. Patients with levels less than 180 ng/ml had a low probability of recovery while levels above 180 ng/ml were associated with a high probability of recovery. Unlike nortriptyline, there is no tight therapeutic window for the efficacy of imipramine.

The correct answer is: Imipramine

Question 73

Not answered

Marked out of 1.00

Flag question

MoK Pharmacokinetics 038

Which of the following truly reflects the effects of smoking?

Select one:

- increases clozapine levels
- no tolerance develops to physical effects
- inhibits CYP1A2
- increases caffeine metabolism
- reduces risk of depression in adults

Your answer is incorrect.

Smoking reduces blood levels of clozapine and is associated with poor therapeutic responses to clozapine. Smokers metabolize caffeine twice as fast as non-smokers. Smoking can substantially increase CYP1A2 activity.

The correct answer is: increases caffeine metabolism

Question 74

Not answered

Marked out of 3.00

Flag question

MoK Pharmacology EMI012

Pharmacodynamics

Select the mechanism of action for each of the following drugs using the given list:

Memantine: Choose...

Buspirone: Choose...

Lofexidine: Choose...

Your answer is incorrect.

Memantine is an antidementia drug that acts through blockade of N-methyl-d-aspartate (NMDA) glutamate receptors. Unlike ketamine that is a high-affinity noncompetitive blocker, memantine is a non-competitive blocker with low affinity and binds only to actively open NMDA channels. Its receptor dissociation rate is relatively fast, and so it does not accumulate and interfere with normal NMDA activity.

Buspirone is a partial agonist on serotonin 5-HT1A receptors. At presynaptic levels, it is mostly a full agonist, which inhibits the release of serotonin, with consequent antianxiety effects. Partial agonist action at postsynaptic receptors appears to account for the antidepressant activity.

Lofexidine is a presynaptic alpha two agonist - leads to reduced central sympathetic tone. Opioid receptors on locus coeruleus projections reduce noradrenergic tone on long-term use. The cellular machinery compensates via up-regulation of adenylate cyclase and maintains normal sympathetic tone in a chronic user. Sudden withdrawal leads to increased adrenergic firing rate (withdrawal symptoms); hence alpha two autoreceptor stimulation that reduces central sympathetic tone helps in treating opioid withdrawal.

The correct answer is: Memantine – NMDA receptor antagonist, Buspirone – 5HT1A partial agonist, Lofexidine – Presynaptic alpha-2 agonist

Question 75
Not answered
Marked out of 3.00
Flag question

MoK Pharmacology EMI014
Adverse reactions
Choose one option from the given list for each description below:

A 22-year-old man is admitted in a secure unit following an assault on two policemen. He is very agitated and dehydrated as a result of the long struggle while arresting him. He was given rapid tranquillisation, soon after which he shows signs of hyperthermia, fluctuant BP and rigidity. — Choose...

A 40-year-old man on lithium treatment has returned from a trip to Greece where he suffered from traveller's diarrhoea. He made a good recovery but when he flew back he noticed tremors and soon developed nausea. — Choose...

A 45-year-old man is on quetiapine. Due to lack of efficacy and poor compliance he is transferred to risperidone depot. During the transition, he was switched to oral risperidone and simultaneous depot administration. He develops a stiff back and some difficulty in rolling over in his bed. — Choose...

Your answer is incorrect.

The first patient has developed Neuroleptic Malignant Syndrome, which is an idiosyncratic reaction, which can occur at anytime during treatment with neuroleptics. It consists of the tetrad of extreme hyperthermia, severe muscular rigidity and confusion, and autonomic fluctuations (varying BP and pulse rate).

The second patient has developed signs of lithium toxicity, which occurs in conditions of overdose or dehydration due to vomiting or diarrhoea. Gastrointestinal symptoms are followed by neurological symptoms. Immediate stopping of lithium and medical attention is required.

The third patient has developed dose-dependent side effects of risperidone due to cumulative drug use leading to extra pyramidal side effects (stiff back is due to axial dystonia).

The correct answer is: A 22-year-old man is admitted in a secure unit following an assault on two policemen. He is very agitated and dehydrated as a result of the long struggle while arresting him. He was given rapid tranquillisation, soon after which he shows signs of hyperthermia, fluctuant BP and rigidity. – Idiosyncratic reaction, A 40-year-old man on lithium treatment has returned from a trip to Greece where he suffered from traveller's diarrhoea. He made a good recovery but when he flew back he noticed tremors and soon developed nausea. – Toxicity, A 45-year-old man is on quetiapine. Due to lack of efficacy and poor compliance he is transferred to risperidone depot. During the transition, he was switched to oral risperidone and simultaneous depot administration. He develops a stiff back and some difficulty in rolling over in his bed. – Dose-dependent side effect

Question 76
Not answered
Marked out of 3.00
Flag question

MoK Pharmacology EMI015
Classifying adverse reactions
Choose one option from the given list for each description below:

A 47-year-old gentleman with a history of chronic opioid misuse is unable to get his drugs for last three days. He develops insomnia, restlessness, diarrhoea and tachycardia — Choose...

A 45-year-old woman with a history of bipolar depression was commenced on Lamotrigine recently. She has developed a severe rash and diagnosed with Stevens- Johnson syndrome. — Choose...

A 67-year-old lady with schizoaffective disorder is on depot antipsychotic for several years. She presents with an odd orofacial movement disorder. — Choose...

Your answer is incorrect.

The first patient suffers from opioid withdrawal effects, related to abrupt discontinuation characterised by insomnia, restlessness, diarrhoea and tachycardia.

In the second case, lamotrigine is known to cause idiosyncratic side effects like Steven Johnson syndrome that is non-dose related.

Tardive dyskinesia and other related ('tardive' - slow onset) syndromes can be classified as time-dependent side effects.

The correct answer is: A 47-year-old gentleman with a history of chronic opioid misuse is unable to get his drugs for last three days. He develops insomnia, restlessness, diarrhoea and tachycardia – Withdrawal effect, A 45-year-old woman with a history of bipolar depression was commenced on Lamotrigine recently. She has developed a severe rash and diagnosed with Stevens- Johnson syndrome. – Idiosyncratic reaction, A 67-year-old lady with schizoaffective disorder is on depot antipsychotic for several years. She presents with an odd orofacial movement disorder. – Time-dependent side effect

Question 77
Not answered
Marked out of 6.00
Flag question

MoK Neurosciences CXEMI012

Cerebral circulation disorders
Identify 3 common symptoms seen in following types of cerebral circulation disorders.

1. Carotid system TIA [ ] [ ] [ ]

2. Vertebrobasilar system TIA [ ] [ ] [ ]

Explanation

Carotid system TIA is also called as anterior circulation TIA. It is characterised by amaurosis fugax, aphasia, hemiparesis, hemisensory loss and hemianopic visual loss due to the involvement of watershed or highly oxygen dependent cortical regions.

Vertebrobasilar system TIA is also called as posterior circulation TIA. It affects brain stem cranial nerve nuclei leading to diplopia, vertigo, vomiting, choking, dysarthria, and ataxia, alexia without agraphia, hemisensory loss, hemianopic visual loss, transient global amnesia, quadriparesis and loss of consciousness (rare).

The correct answer is:
Carotid system TIA - Amaurosis fugax, Hemiparesis, Aphasia,

Vertebrobasilar system TIA - Dysarthria, Ataxia, Diplopia

Question 78
Not answered
Marked out of 3.00
Flag question

MoK Neurosciences EMI018
Neuroimaging
Choose ONE most relevant investigation for each scenario described below.

Differentiation of drug-induced parkinsonism from Parkinson's disease in an elderly woman. — Choose...

A researcher is interested in comparing the degree of dopamine release after cocaine use and amphetamine use using PET scan. — Choose...

Indirect measurement of cerebral metabolism using blood flow measures. — Choose...

Your answer is incorrect.

Beta CIT and FP-CIT (DATscan) are SPECT ligands that bind to dopamine transporters found on nerve endings. In Parkinson's disease where there is a loss of dopaminergic neurons, there is decreased uptake of beta CIT, in contrast to drug-induced Parkinsonism, where the uptake is normal. 11C Raclopride is a PET ligand that binds to D2 dopaminergic receptors. This can be used to determine the amount of dopamine bound to D2 receptors. If there is excess dopamine bound to D2 receptors, there will be less raclopride uptake. Because blood flow to the brain is tightly coupled with local brain metabolism and energy use, the 99mTc-HMPAO is a SPECT tracer used to assess brain metabolism regionally, in an attempt to diagnose causal pathology such as dementia.

The correct answer is: Differentiation of drug-induced parkinsonism from Parkinson's disease in an elderly woman. – ß-CIT SPECT, A researcher is interested in comparing the degree of dopamine release after cocaine use and amphetamine use using PET scan. – 11C raclopride, Indirect measurement of cerebral metabolism using blood flow measures. – Technetium HMPAO SPECT (99mTc).

Question 79

Not answered

Marked out of 3.00

Flag question

MoK Neurosciences EMI020

Neuropeptides

For each of the following descriptions choose ONE correct option from the above list.

Produced in response to fat levels in adipose tissue — Choose...

Produced by stomach and pancreas following food intake — Choose...

Found in hypothalamus and observed to be abnormal in some subjects with obesity — Choose...

Your answer is incorrect.

Leptin is an adipose-derived hormone that circulates at levels proportional to body fat and controls food intake and energy expenditure by acting on the mediobasal hypothalamus.

Ghrelin stimulates hunger and is usually produced by P cells lining the fundus of the stomach and epsilon cells in the pancreas. The levels increase before meals and decrease after meals.

Leptin binds to neuropeptide Y neurons in the arcuate nucleus so that it decreases the activity of neuropeptide Y. The main effect of increased NPY is increased food intake and decreased physical activity.

Kaplan & Sadock Synopsis of Psychiatry 10th edition; pg108.

The correct answer is: Produced in response to fat levels in adipose tissue – Leptin, Produced by stomach and pancreas following food intake – Ghrelin, Found in hypothalamus and observed to be abnormal in some subjects with obesity – Neuropeptide Y

Question 80

Not answered

Marked out of 3.00

Flag question

MoK Neurosciences EMI021

EEG wave forms

Identify the wave forms present during different phases of sleep

REM sleep — Choose...

Stage 2 NREM Sleep — Choose...

Stage 4 NREM Sleep — Choose...

Your answer is incorrect.

REM sleep (25% sleep in adults) is characterized by a high level of brain activity and physiological activity similar to those in wakefulness. EEG shows low-voltage, mixed-frequency (theta and slow alpha) activity similar to the awake state. Sawtooth waves are seen. The REM episodes increase in length as the night unfolds.

NREM sleep: (75% sleep in adults). Most physiological functions are markedly lower than in wakefulness (decreased muscle tone, respiration, temperature and heart rate). It is classified as stages 1 to 4 with increasing amplitude and decreasing frequency of EEG activity. Stage 1 sleep (the drowsy period - 5%) - very light; when awoken from this stage one denies being asleep. Low voltage theta activity with sharp waves. Stage 2 sleep (45% - most common) shows the development of sleep spindles and K complexes. Stage 3 sleep (12% of sleep time) is comprised of <50% delta waves. Stage 4 sleep (13% of sleep time) is comprised of >50% delta waves. Physiological functions are at the lowest during stage 4. Stage 3 & 4 together constitute slow wave sleep (SWS). SWS dominates the initial part of the sleep.

The correct answer is: REM sleep – Saw tooth waves, Stage 2 NREM Sleep – Sleep spindles and K-complexes, Stage 4 NREM Sleep – More than 50% delta waves

Question 81

Not answered

Marked out of 3.00

Flag question

MoK Neurosciences EMI022

Basal Ganglia Disorders

Match the disorders given below with most commonly reported changes in the basal ganglia;

A 10-year-old boy with multiple tics and coprolalia — Choose...

A 67-year-old gentleman with tremors, rigidity and bradykinesia — Choose...

A 46-year-old lady with recurrent obsessional thoughts of contamination and compulsive hand washing repetitively — Choose...

Your answer is incorrect.

Multiple tics with coprolalia in a young boy is suggestive of Tourette's syndrome, which is believed to be caused by striatal dopaminergic dysfunction.

In Parkinson's disease, depigmentation of the substantia nigra is the common finding. Subcortical Lewy bodies are seen. Striatal overactivity may lead to bradykinesia (excessive motor-gating).

In OCD, both reduced and increased volumes of caudate nuclei are reported.Patients have higher caudate blood flow. Increased caudate metabolism has been found to reduce after effective treatment of the OCD (Ring & Serra-Mestres, 2002).

Ring HA and Serra-Mestres J. (2002), Neuropsychiatry of the basal ganglia. Retrieved from http://jnnp.bmj.com/content/72/1/12.full

The correct answer is: A 10-year-old boy with multiple tics and coprolalia – Striatal dopaminergic dysfunction, A 67-year-old gentleman with tremors, rigidity and bradykinesia – Depigmentation of substantia nigra, A 46-year-old lady with recurrent obsessional thoughts of contamination and compulsive hand washing repetitively – Increased caudate blood flow

Finish review